



Glanduliferins A and B, two new glucosylated steroids from *Impatiens glandulifera*, with *in vitro* growth inhibitory activity in human cancer cells

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ABSTRACT

Impatiens glandulifera has been imported from Himalaya in Europe and is considered as an invasive alien plant whose spreading arouses increasing interest among scientific literature. Via anti-cancer bioguiding, two new glucosylated steroids, named glanduliferins A and B, were isolated from the dried stem of *I. glandulifera* plants, together with the well-known α -spinasterol and 2-methoxy-1,4-naphthoquinone, which are also isolated from roots and leaves. They were characterized as 17-(2-hydroxy-2-pentamethylcyclopropyl-ethyl)-10,13-dimethyl-2,3,4,5,6,9,10,11,12,13,14,15,16,17-tetradecahydro-1H-cyclopent[a]phenanthren-3-O-(4-O-acetyl)- α -D-glucopyranoside and 17-(4-ethyl-1,5-dimethyl-hex-2-enyl)-10,13-dimethyl-2,3,4,5,6,9,10,11,12,13,14,15,16,17-tetradecahydro-1H-cyclopent[a]phenanthren-3-O-(6-O-acetyl)- β -D-glucopyranoside using various NMR and HRESIMS techniques and chemical methods. *In vitro* determination of the growth inhibitory activity of the four isolated compounds using the MTT colorimetric assay revealed mean IC₅₀ growth inhibitory value of ~30 μ M for glanduliferin A while glanduliferin B and α -spinasterol were poorly active till 100 μ M. 2-methoxy-1,4-naphthoquinone revealed to be active in the single micromolar digit range as previously described. Quantitative videomicroscopy analyses of the effects of the effects of glanduliferins A and B suggested cytostatic rather than cytotoxic activity in U373 glioblastoma (GBM) cells.

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1. Introduction

Cancer is a serious health problem not only in developed countries but also in developing countries [1,2]. Among all the weapons used by oncologists to combat cancer, natural sources have provided more than 50% of marketed anticancer drugs since the birth of the modern anticancer chemotherapy during World War II [3–5]. However, several major problems have been identified during the last decades as actual barriers to efficiently combat cancer. Among these problems remains the fact that most of natural products are cytotoxic and induce the multidrug resistance (MDR) phenotype in cancer cells [6,7]. Most natural products are also pro-apoptotic, while cancers associated with dismal prognoses, such as gliomas [8], melanoma [9] and metastatic cancers [10,11] resist to pro-apoptotic stimuli.

In this line, the present study deals with the search of new potential anticancer drugs from the plant *Impatiens glandulifera*, which is an annual herbaceous plant 1–2 m high, very vigorous, with erect, robust,

translucent stems, reddish and swollen at the nodes of intersection [12]. This plant originated from western Himalaya and it was first introduced in Europe as an ornamental plant in 1839 [13]. Since, it is widespread in most Northern European countries, thus *I. glandulifera* has increasingly invaded deciduous and coniferous forests disturbed by wind throws and/or intensive forest management [14]. Now, *I. glandulifera* is one of the 100 worst invasive alien plants in Europe; its planting is prohibited in several countries and ecological control is urgently needed. *Impatiens* species, commonly named jewelweeds, have been shown to contain various chemicals including among others naphthoquinones [15–17], glucoside naphthalenes [18], steroids [15], baccharane glycosides [19–22] and saponins [23]. The first isolation of a bioactive compound from *I. glandulifera* was reported in 1974 by Chapelle who isolated 2-methoxy-1,4-naphthoquinone (2MNQ) and lawsone from dried leaves [24]. This was further confirmed by [16] who showed that *I. glandulifera* content in 2MNQ is higher than other *Impatiens* species. Later, [25] reported again 2MNQ isolation from *I. glandulifera* as an anti-cancer compound against hepatocellular carcinoma cells. In this study, we thus collected senescent plants in September/October when its 2MNQ content is low [26] in order to facilitate the identification of potential new anti-cancer bioactive compound(s).

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2. Experimental

2.1. General

^1H and ^{13}C NMR spectra were recorded at 400/100 MHz in CDCl_3 , unless otherwise noted, on Bruker spectrometers. The same solvent was used as internal standard. Carbon multiplicities were determined by DEPT spectra [27]. DEPT, COSY-45, HSQC, HMBC and NOESY experiments [27] were performed using Bruker microprograms. HRESI and ESI, and APCIMS spectra were recorded on Agilent Technologies 1100 LC/MSD TOF and Agilent Technologies 6120 Quadrupole LC/MS instruments, respectively. Analytical and preparative TLCs were performed on silica gel (Kieselgel 60, F254, 0.25 and 0.50 mm respectively, Merck, Darmstadt, Germany) plates. The spots were visualized by exposure to UV light (254 nm) and/or by spraying first with 10% H_2SO_4 in MeOH and then with 5% phosphomolybdic acid in EtOH, followed by heating at 110 °C for 10 min and/or by exposure to iodine vapors. Column chromatography was performed using silica gel (Kieselgel 60, 0.063–0.200 mm, Merck, Darmstadt, Germany).

2.2. Plant material

I. glandulifera were collected along the Semois river in Belgium (Bohan, Belgium, N 49°52'47" E 4°53'47") when it was senescent, i.e. in September/October. Plants were manually unearthed. Separation of the roots, stem, leaves and flowers with seeds was conducted before air drying.

2.3. Extraction and purification of glanduliferins A (1) and B (2)

The dry powders (10 g) of the different air-dried *I. glandulifera* parts (roots, stems and leaves) were extracted twice with *n*-hexane, EtOAc and MeOH (1 × 100 ml) in shaking conditions for 4 h. The extracts were filtered through a paper in a Buchner funnel and evaporated under reduced pressure. The organic extracts were combined, dried (Na_2SO_4) and evaporated under reduced pressure. All the organic extracts were tested by means of the MTT colorimetric assay on three human cancer cell lines in order to process with a bioguiding-approach for the identification of new potential anticancer compounds. As MeOH and *n*-hexane extracts were not or poor active respectively, we focused on the EtOAc extract. Thus, EtOAc organic extract (555.0 mg) obtained from stems (1 kg) showing a mean IC_{50} concentration of 33 $\mu\text{g}/\text{ml}$ on three cancer cell lines was bio-guided purified by CC eluted with $\text{CHCl}_3/\text{i-PrOH}$ (95:5 v/v) affording 9 groups of homogeneous fractions (labeled I.G.S 1–9). The active fraction IG S3 (33.1 mg) was purified on TLC eluted with *n*-hexane/ Me_2CO (7:3 v/v) obtaining 4 fractions. The second one resulted to be a pure metabolite appearing as white solid (Rf 0.67, 9.1 mg, 9.1×10^{-4} %) identified as α -spinasterol (3). The active fraction IG S2 (48.9 mg) was purified on TLC eluted with *n*-hexane/ Me_2CO (7:3 v/v) obtaining a major metabolites (Rf 0.36, 8.1 mg) identified as 2-methoxy-1,4-naphthoquinone (2MNQ; 4). The fraction IG S8 (71.5 mg) was purified on TLC eluted with $\text{CHCl}_3/\text{i-PrOH}$ (9:1 v/v) obtaining one main band named S8.5. This latter was further purified by two successive steps on TLC eluted with *n*-hexane/ Me_2CO (1:1 v/v) and EtOAc/*n*-hexane (1:1 v/v) yielding two homogeneous oily metabolites named glanduliferin A (1, 3.7 mg) and glanduliferin B (2, 1.2 mg) and characterized as below reported as new α - and β -glucosylated steroids. α -Spinasterol and 2MNQ (3 and 4) were also isolated from the EtOAc extract of roots and leaves of the same plant.

Glanduliferin A (1): oil; Rf 0.62, EtOAc/*n*-hexane (1:1 v/v); ^1H and ^{13}C NMR, see Table 1; HRESIMS (+) m/z 650.4695 [$\text{M} + \text{NH}_4$] $^+$ (calcd for $\text{C}_{37}\text{H}_{64}\text{NO}_8$ 650.4632), 1282.9044 [$2\text{M} + \text{NH}_4$] $^+$ (calcd for $\text{C}_{74}\text{H}_{124}\text{NO}_{16}$ 1282.8920); ESIMS (+) m/z 655 [$\text{M} + \text{Na}$] $^+$, 1287 [$2\text{M} + \text{Na}$] $^+$.

Glanduliferin B (2): oil; Rf 0.77, EtOAc/*n*-hexane (1:1 v/v); ^1H and ^{13}C NMR, see Table 1; HRESIMS (+) m/z 634.4744 [$\text{M} + \text{NH}_4$] $^+$ (calcd

Table 1
 ^1H and ^{13}C NMR data of glanduliferins A and B (1 and 2)^a.

Position	1			2		
	δ_{C}^b	δ_{H} (J in Hz)	HMBC	δ_{C}^b	δ_{H} (J in Hz)	HMBC
1	37.9	1.85	H-2A, H-2B, Me-18	37.1	1.84 m	
2	34.0 ^c	1.86 m	Me-18	29.4	1.07 m	
		1.46 m				
3	80.5	3.97 td (5.2)	H-1', Me-18, H ₂ -4	79.2	3.64 m	H-1'
4	45.4	2.03 m		38.0	2.02 (2H) m	
		1.42 m				
5	41.4	1.62 m	H-6A	40.2	2.04 m	
6	35.5	1.56 (2H) m	H-7	29.7	1.77 (2H) m	
7	118.3	5.24 br d (6.1)	H-11	117.2	5.15 m	H-14
8	139.7		H-6A, H ₂ -11	139.2		
9	47.7	1.02 m	H ₂ -11, H-12A, H ₂ -6, H-5, Me-18	49.4	1.65 m	
10	39.3			34.4		
11	21.5	1.60 m		21.5	1.56 m	
		1.64 m			1.46 m	
12	40.6	1.19 m	H-14	39.4	1.24 (2H) m	
		1.99 m				
13	44.6		H-15	43.1		
14	52.9	1.60 m	Me-19	55.0	1.83 m	H-7
15	27.6	1.60 m		23.0	1.42 m	
		1.56 m			1.30 m	
16	28.2 ^c	1.02 m	H-17	28.5	1.91 (2H) m	
		0.96 m				
17	54.4	0.75 m	H-12A, H ₂ -16, Me-19	55.8	1.27 m	H-21, H-20
18	9.3	1.05 s	H ₂ -1	12.0	1.04 s	
19	16.5	0.95 s	H ₂ -20	13.0	0.55 s	
20	34.0 ^c	1.57 (2H) m	Me-19	40.8	2.04 m	H-21, H-23
						H-17, H-22
21	78.1	3.20 dd (9.9, 5.4)	H-17, H ₂ -16, H ₂ -20	21.4	1.02 d (6.5)	H-21, H-24
22	34.0 ^c		Me-25, H-21	138.7	5.15 dd (15.0, 8.5)	
23	47.9		Me-28, Me-29	129.4	5.02 dd (15.0, 8.5)	
24	43.4		Me-26, Me-27	51.2	1.54 m ^d	H-22
25	28.2 ^d	0.99 s		31.8	1.54 m ^d	
26	15.0	0.77 s		21.1	0.85 d (6.5)	
27	18.7	0.76 s		19.0	0.81 d (6.5)	
28	18.2	1.18 s		25.4	1.18 m	
29	22.1	0.93 s		12.3	0.80 t (6.5)	
1'	102.9	4.66 d (3.7)	H-3, H ₂ -6'	101.0	4.40 d (8.5)	H-3, H ₂ -6'
2'	70.5	3.53 m		73.9	3.36 t (8.5)	H-1'
3'	71.4	3.71 m		73.0	3.43 m	H-2', H-4'
4'	72.5	5.17 t (10.9)		75.9	3.58 t (8.5)	H-5'
5'	69.3	3.76 m		71.8	3.41 m	
6'A	62.4	4.11 dd (12.4, 2.9)		63.0	4.26 dd (12.3, 1.0)	H-1'
6'B	62.4	3.45 dd (12.4, 4.8)		63.0	4.51 dd (12.3, 4.1)	H-1'
COMe	170.3		H-4', MeCO	171.0		MeCO
MeCO	22.2	2.02 s		20.1	2.11 s	

^a 2D $^1\text{H}, ^1\text{H}$ (COSY) and 2D $^{13}\text{C}, ^1\text{H}$ (HSQC) NMR experiments delineated the correlations of all protons and the corresponding carbons.

^b Multiplicities were assigned by DEPT spectra.

^c These signals are overlapped.

^d These signals were overlapped.

for $\text{C}_{37}\text{H}_{64}\text{NO}_7$ 634.4683), 1250.9150 [$2\text{M} + \text{NH}_4$] $^+$ (calcd for $\text{C}_{74}\text{H}_{124}\text{NO}_{14}$ 1250.9022); ESIMS (+) m/z 639 [$\text{M} + \text{Na}$] $^+$; APCIMS (+) m/z 607 [$\text{M} + \text{H}$] $^+$.

2.4. Determination of sugar moieties of glanduliferins A (1) and B (2)

For compositional analysis, glanduliferins A and B were hydrolyzed with 0.1 M HCl (100 °C, 48 h). Monosaccharides were identified as alditol acetate by GLC using a Hewlett-Packard 5880 instrument

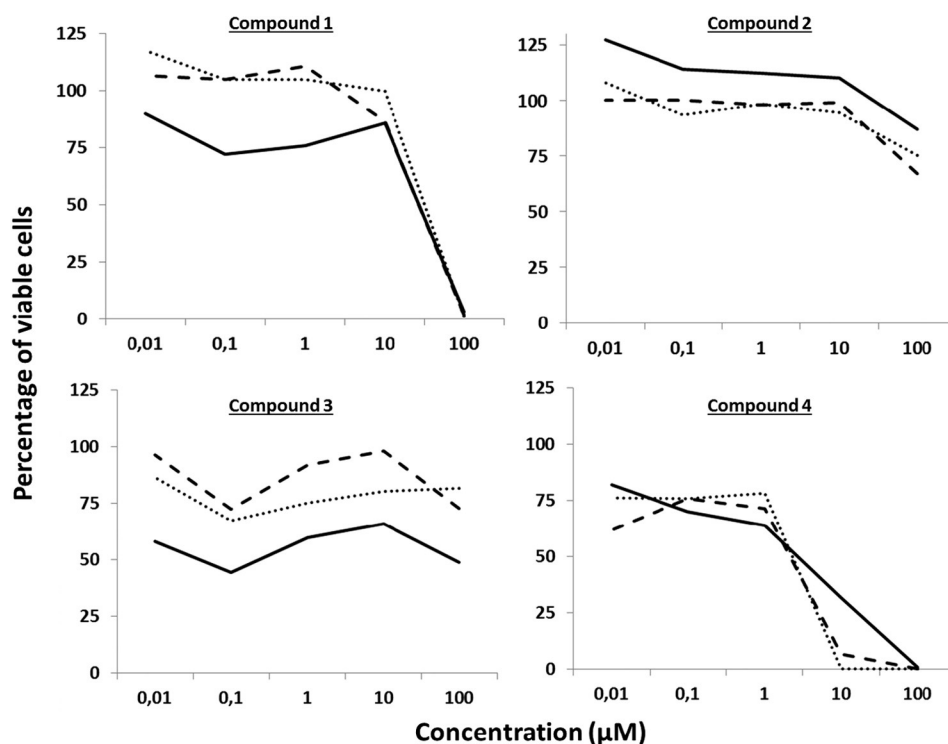


Fig. 2. MTT response curves of glanduliferins A and B (**1** and **2**), α -spinasterol and 2-methoxy-1,4-naphthoquinone (**3** and **4**). Hatched bars represent the A549 human non-small cell carcinoma cell line, the dotted bars the SKMEL-28 human melanoma cell line and the continuous line the U373 human glioma cell line. Curves represent the mean results of six replicates conducted once at the same time.

group and H-4'. The chemical shifts of other protons and carbons of the glucosyl residue were assigned by the correlations observed in the COSY and HSQC spectra [27] according to the literature [33,34]. The ^1H and ^{13}C NMR spectra showed also the signal pattern of a 3-hydroxy steroid as those of the hydroxylated methine C-3 at δ 80.5 (d)/3.97 (td, $J = 5.2$ Hz), of the double bond between C-7 and C-8 at δ 118.3 (br d, C-7)/5.24 (d, $J = 6.1$ Hz) and 139.7 (s, C-8) which were assigned on the basis of the couplings observed in the COSY and HSQC spectra. Similarly, the signals at δ 1.62 and 1.60 (both m) were assigned to the two protons (H-5 and H-14) and those at δ 1.05 and 0.95 (both s) to the two methyls (Me-18 and Me-19) bonded to the head-bridge carbons C-10 and C-13 of the A/B and C/D ring junctions which resonated at δ 39.3 (s, C-10) and 44.6 (s, C-13). The other protonated head-bridge carbons showed signals at δ 41.4 (d, C-5) and 52.9 (d, C-14). Other two methines (HC-9 and HC-17) of the cyclopentanoperhydrophenanthrene system were observed at δ 44.7 (d, C-9)/1.02 (m) and 54.4 (d, C-17)/0.75 (m). All these attributions were confirmed by the couplings observed in the HMBC spectrum (Table 1).

The side chain at C-17 contains the residual ten carbons including one hydroxylated methine, one methylene and five methyl groups as well as three quaternary carbons and the resulting unsaturations. The $\text{H}_2\text{C}-20$ at δ 34.0 (t)/1.57 (2H, m) resulted to be the head of this side chain, as resulted from the couplings observed in the HMBC spectrum between C-20 and Me-19 and C-21 with H_2-20 . The protons of this methylene group ($\text{H}_2\text{C}-20$) coupled in the COSY spectrum only with the proton of the hydroxylated secondary carbon resonating at δ 78.1 (d, C-21)/3.20 (m). The latter (H-21) in the HMBC spectrum coupled with the quaternary carbon (C-22) at δ 34.0 and this in turn with the geminal Me-25 resonating at δ 28.2 (q)/0.99, (s). C-22 is one of three quaternary carbons of the hexasubstituted cyclopropane ring, which represents the tail of the side chain residue. The other two quaternary carbons C-23 and C-24, observed at δ 47.9 and 43.4, were assigned on the basis of their couplings in the HMBC spectrum with the adjacent methyl group Me-28 and Me-29 and Me-26 and Me-27

observed at δ 18.2 (q, C-28)/1.18 (s) and 22.1 (q, C-29)/0.93 (s) and 15.0 (q, C-26)/0.77 (s) and 18.7 (q, C-27)/0.76 (s).

On the basis of these results glanduliferin A (**1**) appeared to be a new α -glucosylated steroid and could be formulated as 17-(2-hydroxy-2-pentamethylcyclopropyl-ethyl)-10,13-dimethyl-2,3,4,5,6,9,10,11,12,13,14,15,16,17-tetradecahydro-1H-cyclopent[*a*]phenanthren-3-O-(4-O-acetyl)- α -D-glucopyranoside. The structure assigned to **1** was supported by further couplings observed in the HMBC spectrum for the phenanthrene cyclopentane ring system. In particular those between C-1 and C-2 with Me-18, C-6 with H-7, C-10 with H-5 and Me-18, C-12 with H-14, C-13 with H-15, C-14 and C-17 with Me-19, C-17 with H_2-16 , C-18 with H_2-1 and C-19 with H_2-20 . The structure assigned to **1** was confirmed by the data of the HRESIMS and NOESY [27] spectra. In fact, the HRESIMS showed the ammonium cluster $[\text{M} + \text{NH}_4]^+$ and the corresponding dimer $[2\text{M} + \text{NH}_4]^+$ at m/z 650.4695 and 1282.9044, respectively. ESIMS recorded in positive ion mode showed the sodiated cluster $[\text{M} + \text{Na}]^+$ and the corresponding dimer $[2\text{M} + \text{Na}]^+$ at m/z 655 and 1287, respectively. In particular, the NOESY correlation observed between H-3 (β -equatorial) at δ 3.97 with H-2 and H-4 (both β -axial) at δ 1.46 and 1.42 and Me-18 (β -axial) at δ 1.05 as well as the multiplicity and constants measured in the ^1H NMR spectrum for the coupling of H-3 with the protons of its adjacent methylene groups ($\text{H}_2\text{C}-2$ and $\text{H}_2\text{C}-4$) allowed to assign the relative configuration to C-3 as depicted in **1**. The lacking of correlation of

Table 2

IC₅₀ growth inhibitory concentrations determined by means of the colorimetric MTT assay over a 72 h period of exposure to each compound (μM).

Compound	A549	SKMEL-28	U373	Mean $\pm$ SEM
1	27	29	31	29 $\pm$ 1
2	>100	>100	>100	>100
3	>100	>100	ND	ND
4	3	2	3	3 $\pm$ 0.2

ND: not determined according to the shape of the curve (see Fig. 2).

Me-18 and H-5 allowed assigning a *trans* configuration at the junction to the A/B ring. The correlations observed in the NOESY spectrum also between both H-3 and H-21 with both Me-27 and Me-28 suggested a bended structure in which the side chain point toward the ring A. This allowed assigning α -configuration to H-17. Finally on the basis of the lacking of correlations between this latter proton and Me-19 and also between this latter with H-14, confirmed that the side chain at C-17 has a β -configuration, and a *trans* configuration could be also assigned to the C/D ring junction. This structure appears very unusual in the group of saponins. However, investigation of a Dreiding model of the side chain of glanduliferin A, indicated no distortions of bond angles and no steric hindrance. NOESY correlations were not observed to assign the relative configuration to C-21, which remain undetermined.

The side chain at C-17 of glanduliferin A is very unusual among the steroids group. A number of steroids with unusual side chains at C-17 were reported in literature [35–44] but only for some of them the biosynthetic pathway was proposed or determined through the use of suitable experimental studies [41–44]. Biosynthetic studies of glanduliferin A are outside the scope of the current work. However,

the most significant data supporting the structures are available as supplementary data.

Glanduliferin B (**2**) has a molecular formula of $C_{37}H_{60}O_7$, as was deduced from its HRESIMS spectrum consistent with a hydrogen deficiency index of eight. The investigation of its 1H and ^{13}C NMR spectra (Table 1) confirmed the presence of the glucose residue which resulted to be linked at C-3 as indicated by the couplings observed in the HMBC spectrum between C-3 with H-1' and C-1' and H-3. The glucosyl residue showed to be β -linked to C-3 for the typical chemical shift of the anomeric proton and carbon observed at δ 101.0 (d, C-1')/4.40 (d, $J = 8.5$ Hz) and it appeared acetylated at C-6' for the expected downfield shift of H₂-6' resonating as two double doublets ($J = 12.3$ and 1.0 and 12.3 and 4.1 Hz) resonating at δ 4.26 and 4.51 while the signals of the acetyl group appeared at δ 171.0 (s, COMe) and 2.11/20.1 (s and q, COMe) [33,34]. The chemical shifts of the other protons and carbons of the glucosyl residue were assigned by the correlations observed in the COSY and HSQC spectra and in agreement to literature data [33, 34]. The 1H and ^{13}C NMR spectra also showed that the steroid moiety of **2** differed from that of **1**. While the signal pattern of a 3-hydroxy colestane steroid as **1** showed a trisubstituted double bond between

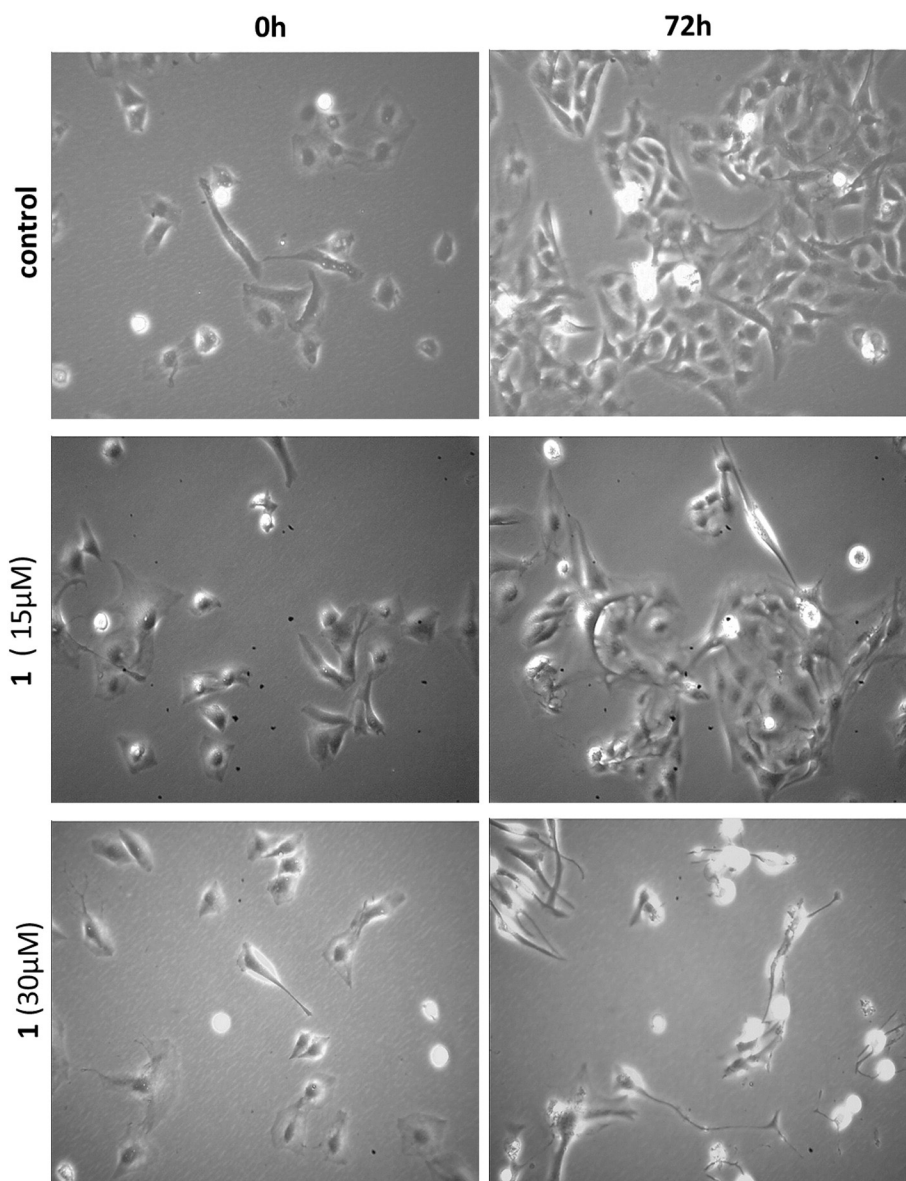


Fig. 3. Illustrations of videomicroscopy results of U373 cells left untreated or treated with glanduliferin A (**1**) at 15 and 30 μ M respectively.

C-7 and C-8 and two protons and the two methyl groups at the head-bridge carbon of the A/B and C/D ring junctions were observed at δ 117.2 (d, C-7)/5.15 (m) and 139.2 (s, C-8), 40.2 (d, C-5)/2.04 (m) and 34.4 (s, C-10), 55.0 (d, C-14)/1.83 (m) and 43.1 (s, C-13), and 12.0 (q, C-18)/1.04 (s) and 13.0 (q, C-19)/0.55 (s), the signals of the side chain appeared to be different. In fact, the side chain at C-17 appeared to be a 1-methyl-4-isopropyl-3-hexenyl, whose signals, attributed on the couplings observed in the COSY and HSQC spectra, resonated at δ 138.7 (d, C-22)/5.15 (dd 15.0, 8.5 Hz) and 129.4 (d, C-23)/5.02 (dd 15.0, 8.5) for the *trans* disubstituted double bond, at δ 40.8 (d, C-20)/2.04 (m), 51.2 (d, C-24)/1.54 (m) and 31.8 (d, C-25)/1.54 (m) for the three methines, δ 21.4 (q, C-21)/1.02 (d, 6.5) for a secondary methyl, δ 21.1 (q, C-26)/0.85 (d, 6.5) and 19.0 (q, C-27)/0.81 (d, 6.5) for the isopropyl, as well as δ 25.4 (t, C-28)/1.18 (m) and 12.3 (q, C-29)/0.80 (t, $J = 6.5$) for the ethyl group [33,34]. On this basis the aglycone of glanduliferin B (**2**) appeared to be the α -spinasterol (**3**). Also by the correlation recorded in the COSY, HSQC and HMBC spectra (Table 1) the protons and carbons chemical shifts of this aglycone were completely assigned for the first time as reported in Table 1, and were in agreement with the partial assignment previously reported for α -spinasterol when its 3-glucosyl was isolated from *Prunella vulgaris* [45]. So that glanduliferin B (**2**) could be formulated as 17-(4-ethyl-1,5-dimethylhex-2-enyl)-10,13-dimethyl-2,3,4,5,6,9,10,11,12,13,14,15,16,17-tetradecahydro-1H-cyclopenta[a]phenanthren-3-O-(6-O-acetyl)- β -D-glucopyranoside. The structure assigned to glanduliferin B (**2**) was confirmed by the data obtained in its HRESIMS which showed the ammonium cluster $[M + NH_4]^+$ and the corresponding dimer $[2M + NH_4]^+$ at m/z 634.4744 and 1250.9150, respectively. ESIMS recorded in positive ion mode showed the sodiated form $[M + Na]^+$ at m/z 639, while its APCIMS spectrum showed the pseudomolecular ion $[M + H]^+$ at m/z 617. The NOESY spectrum showed the significant correlations between H-3 with H1', H-7 with H2-6, H-20 with H-22, H-23 with H-24 and H-1' with H-3'. The lacking of correlation between Me-18 and Me-19 with H-5 and H-14 respectively confirmed the *trans* configuration assigned to both the junction of A/B and C/D rings. Finally the aglycone obtained by acid hydrolysis of glanduliferin B (**2**) showed the same chromatographic behavior of α -spinasterol also in mixture by TLC on direct and reverse phases, eluted with different solvents, and the same 1H NMR and ESIMS spectra when recorded in the same conditions [46]. These results, which are consistent with the above described NOESY correlations, allowed to assign the stereochemistry of glanduliferin B in agreement with that reported in literature for α -spinasterol [47] and as depicted in Fig. 1.

The *in vitro* evaluation of the growth inhibitory activity of the EtOAc extracts revealed no anti-cancer effects from the seeds while the stem, the leaves and the roots extracts displayed mean IC_{50} growth inhibitory concentrations ranging from 33 to 66 $\mu g/ml$ on the three cancer cell lines under study. Fractionation led to increased activity with IC_{50} values ranging between 12 and 22 $\mu g/ml$. Despite that senescent *I. glandulifera* plant has low levels of 2MNQ (**4**) [26], this compound was isolated from the three parts of the plant and displayed anti-cancer activity *in vitro* in the single digit micromolar range on the three cell lines (Fig. 2, Table 2). 2MNQ induces apoptosis or necrosis of gastric cancer cells at concentrations of 25 and 50 μM via superoxide production [48]. IC_{50} concentration of 2MNQ on HepG2 cells was 34 μM [25] but at lower concentration 2MNQ was also shown to decrease migration and invasion of breast cancer cells *in vitro* [49].

α -Spinasterol (**3**) was isolated from stem extract fraction. Its growth inhibitory activity has also been evaluated using the MTT colorimetric assay (Fig. 2, Table 2). No clear dose-dependent response was observed and its IC_{50} growth inhibitory concentration was not reached at least till 100 μM in A549 non-small-cell lung cancer (NSCLC) and SKMEL-28 melanoma cell lines (Fig. 2). U373 glioblastoma (GBM) cells appeared to be slightly sensitive to **3**. Previous study reported similarly weak activity from 20 to 80 μM for both α -spinasterol (**3**) and spinasterol [50]. This latter exerts also weak anti-microbial activity in the range of 20–

80 $\mu g/ml$ and could affect steroid-dependent cancer models via estrogen receptor activity modulation [51].

Glanduliferins A (**1**) and B (**2**) were isolated from an active fraction of the stem extract. The evaluation of their *in vitro* growth inhibitory activity in three human cancer cell lines (U373 GBM, A549 NSCLC and SKMEL-28 melanoma cells) revealed mean IC_{50} growth inhibitory values ranging from ~30 μM and >100 μM , respectively (Table 2). At 100 μM , glanduliferin B (**2**) inhibited the cancer cell growth by only 15 to 25% while no cell remained viable with glanduliferin A (**1**) at that concentration after 72 h of incubation (Fig. 2). We therefore compared **1**- and **2**-related effects on U373 GBM cells by means of videomicroscopy. At its IC_{50} growth inhibitory concentration (as obtained by means of the MTT colorimetric assay; Table 2) compound **1** completely blocked U373 cell growth as quantitatively reported in Fig. 3. Compound **1** also led to altered cell morphology and even cytotoxicity as illustrated in Fig. 4. At half of its IC_{50} growth inhibitory concentration compound **1** exerted pure cytostatic activity (Figs. 3 and 4). In contrast, compound **2** displayed weak cytostatic effects with 25% of growth inhibition after 72 h of treatment at 100 μM (Fig. 4B). Considering that α -spinasterol and its glucoside have any and reduced activity, the higher activity exerted by glanduliferin A could be due to its different side chain at C-17, which as above reported is bended on the tricyclic phenanthrene ring pointing toward A ring. Probably also the opposite stereochemistry at C-3 and the different acetylation of glucose residue could play a role in the activity.

In conclusion, two new glucosylated steroids named glanduliferins A and B, belonging both to a cholestane subgroup, were isolated from *I. glandulifera*. While glanduliferin B showed a structure close to the other steroids of the same subgroup [32], and in particular to 3- β -glucosylated- α -spinasterol, glanduliferin A appeared original for the side chain attached to the cyclopentaperhydrophenanthrene system,

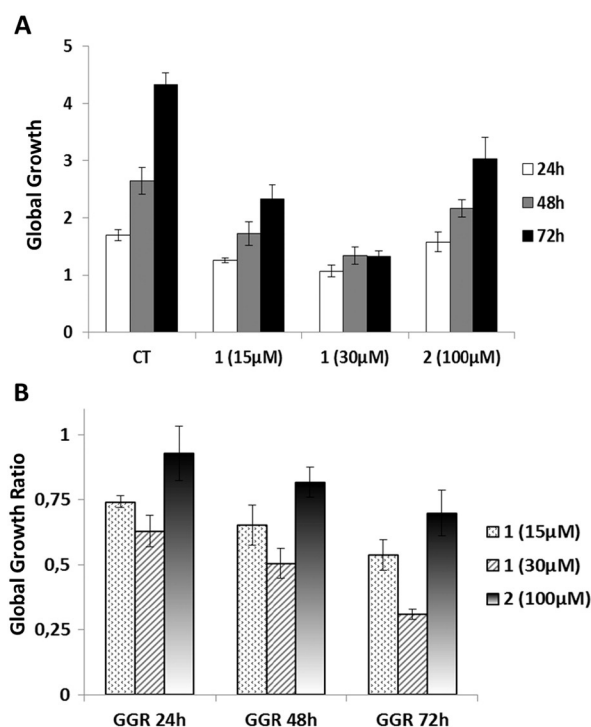


Fig. 4. Quantitative data of the videomicroscopic analysis conducted on U373 cells with glanduliferins A and B (**1** and **2**). A: Global growth analysis (GG): the global growth is the ratio between the number of cells at $T = x$ and the number of cells at $T = 0$. A global growth of 2 means that the population has doubled at the time investigated. B: the global growth ratio (GGR) is the ratio between the treated global growth and the control global growth. A GGR of 0.5 means that the global growth has been reduced by 50% at the time investigated. Results are expressed as mean $\pm$ SEM of three replicates.

for its length and also for the novelty of its tail represented by a 1,2,2,3,3-pentamethylsubstituted cyclopropyl. The originality we claim for compound **1** must be analyzed with respect to those data reported in literature for a large number of steroids belong to different subgroups [32]. Indeed, glanduliferins A and B belong to the saponins group of natural compounds, glycosylated triterpenoids, steroids or steroidal alkaloids, found in many plant species. Some of them showed the capacity to inhibit the growth of tumor cells *in vitro* [52]. Glanduliferin A exerts cytostatic anti-cancer effects *in vitro* at 15 μ M, with a mean IC₅₀ of ~30 μ M, while its analogs glanduliferin B and α -spinasterol did not reach 50% of growth inhibition till 100 μ M on the three cancer cell models under study.

Conflict of interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.fitote.2015.12.016>.

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